The Covid-19 Impact on Temporal Threshold Networks in the US Stock Market

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The stock market is a large, complex, interrelated system with unpredictable dynamical behaviour. Network science offers a powerful framework for modelling such complex systems. The Covid-19 pandemic created a new, uncertain economic environment, with stock crashes unseen since the Great Depression. This study reveals the hidden impacts of Covid-19 on stock market structure and stock similarity dynamics. The temporal, time-varying networks were built using stocks from the entire NASDAQ stock exchange, a total of 2456 stocks for the period from October 2019 to October 2021. Pairwise correlations of logarithmic returns were used to construct time-varying threshold networks, with a criterion proposed for selecting a suitable threshold based on the maximum edge weight of the Minimum Spanning Tree (MST). The results reveal that the Covid-19 pandemic caused a temporary collapse in market diversification, unifying all sectors into a highly interconnected structure and reducing community modularity. Low volatility financial and technology stocks became the most central in the network during the crisis. As volatility subsided, the networks gradually regained their pre-pandemic hierarchical structure. These findings highlight the value of network-based approaches for monitoring systemic risk and indicators of market recovery.

I. INTRODUCTION

Network science is becoming an increasingly powerful tool to model large complex systems [1]. It has been applied in many fields like biomedical science for locating important genes [2], social media for friend suggestions [3], and epidemiology in predicting the spread of contagious diseases [4]. Mantegna [5] constructed stock networks based on the correlation matrix of the S&P 500 index. This seminal paper demonstrated the utility of network science for extracting community structure from complex stock networks, inspiring future research in different markets [6–10]. In these studies, individual stocks are represented by nodes of a network. Edges, connecting these stock pairs, are weighted by a distance that describes the correlation of their price changes between two stocks, where lower distances signify greater similarity of price co-movement [11].

The original complete networks, where any two stocks are connected by a weighted edge, are often too large to handle. Authors have proposed different methods to reduce the number of edges [5, 10, 12–14]. These constructions attempt to retain the most similar edges, but may suffer from extreme information loss [15]. Boginski et al. [14] propose a threshold network that solves this issue, especially for large networks, by filtering edges above a certain threshold. This approach, however, raises the question of how the threshold should be chosen [16]. The current study creates threshold networks based on a proposed criterion that is practical for large networks, ensures that the threshold network is connected and makes the network's organizational structure clearly visible. In our case, the networks depend on time (temporal networks), which means that a threshold must be calculated independently for hundreds of individual networks. We propose to use the maximum edge weight that occurs in the Minimum Spanning Tree (MST) as the threshold. This prescription is efficient and easy to automate, and we demonstrate that changes in market structure are clearly reflected in the threshold networks. It does not depend on the nature of the data that is analyzed and can therefore be applied across diverse fields of study.

The threshold networks allow one to determine the community structure of the network, i.e., to identify stocks that are affected in a similar manner by external factors. These results can be useful in portfolio optimisation and diversification. For instance, risk can be reduced by investing in different communities influenced by different factors [17]. The networks also allow a study of the most central stocks, which are highly associated with the overall stock market movement [7]. This may enable researchers to narrow down the factors affecting the wider market by analysing potential causes of price fluctuations in the most central stocks.

Covid-19 is an example of a crisis that devastatingly affected the financial markets, particularly stock prices. In March 2020, the World Health Organization declared the pandemic, causing large company stock prices to plummet by over 30% in just weeks [11, 18]. The crash marked one of the fastest price drops in history, not witnessed since the Great Depression, with spikes of volatility that set new records [18, 19]. The pandemic's sheer magnitude resulted in extreme financial market chaos for all economies, only comparable to the 2008 Global Financial Crisis and the Great Depression a century ago [20]. Although the consequences were vast, the Covid-19-induced financial market downturn and economic recession were the shortest in history, lasting just two months [21]. The Covid-19 pandemic created an entirely new economic environment, thus, confronting researchers with the challenge to analyze and explain such an anomalous event.

Several analyses revealed that stock networks un-

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dergo topological changes during periods of market crisis [10, 22]. The 2008 financial crisis increased similarity across stocks, increased network density, and disrupted sectoral community patterns [9, 22, 23]. Heiberger et al. [22] built threshold networks to analyse the dot-com and 2008 financial crises and observed that, before the crisis, the networks had well-partitioned communities that coincided roughly with the sector. However, as crisis hits, this community structure merges into bigger, less well-defined communities. This is due to an increased complexity of the network as similarity between stocks increases. Noticing the return of a stable community structure of the networks post-crisis can provide investors with a signal of market recovery [3, 10].

Few studies have yet analyzed Covid-19's impact on temporal threshold networks [24, 25]. This study found that during the Covid-19 crisis, low-volatility financial stocks became the most central nodes in the network. The well-defined community structure that existed prior to the crisis temporarily collapsed into less distinct groups, reflecting a breakdown in sectoral differentiation and a loss of market diversity. These findings enhance understanding of stock market dynamics during periods of systemic stress and recovery.

The remainder of the paper is structured as follows: Section II contains information about data collection, and the construction of the networks. A practical criterion for the selection of a suitable threshold distance is introduced and justified here. Section III contains information about communities in the network, tracking the key communities over time. Section IV identifies central stocks in the network and describes how their attributes, like sector, etc., change over time. Section V contains a summary of the important findings and concluding remarks.

II. DATA COLLECTION AND NETWORK CONSTRUCTION

A. Data collection

Daily price data spanning 2 years, between October 2019 and October 2021, was obtained for 2456 stocks by using the Python yfinance library to scrape data from Yahoo Finance. The time frame was carefully selected to present the effect of Covid-19, with a period of market stability 6 months before the onset of the Covid-19 pandemic (Late February 2020) and a period after to depict recovery. This study includes all stocks from the NASDAQ for which price data was available for the entire period of the study. Any stock that did not date back sufficiently 2 years or had missing price data was removed during the data collection process.

Instead of simply the increase (or decrease) in price of a stock from one business day to the next, logarithmic returns were taken. The **logarithmic return** $R_i(t)$ of stock i between times $t-\Delta t$ and t is given by [26]

$$R_i(t) = \ln P_i(t) - \ln P_i(t-\Delta t) = \ln \frac{P_i(t)}{P_i(t-\Delta t)}, \quad (1)$$

where $P_i(t)$ is the daily adjusted closing price for stock i on day t and Δt is an increment scale, taken to be 1 trading day to represent the daily logarithmic return increment.

Categorising stocks according to different attributes is an important tool for the economic interpretation of the stock market data. The attributes used in this study are the stock company sector and volatility levels. The proportion of the different sectors is summarised in Table I.

B. Similarity Measures

Loosely speaking, two stocks are considered to behave similarly in a given time-frame, if the prices of both increase or decrease at the same rate and time. This notion is made precise by the **Pearson correlation coefficient** $p_{ij}(\tau)$ between the logarithmic returns of stocks i and j for the period τ, which is given by [27]:

$$p_{ij}(\tau) = \frac{\langle R_i(\tau)R_j(\tau)\rangle - \langle R_i(\tau)\rangle \langle R_j(\tau)\rangle}{\sqrt{\left\langle R_i(\tau)^2 - \langle R_i(\tau)\rangle^2\right\rangle \times \left\langle R_j(\tau)^2 - \langle R_j(\tau)\rangle^2\right\rangle}} \quad (2)$$

where angular brackets indicate that a quantity is averaged over the time period. We choose a time period τ of 21 days, which is a typical number of trading days in a calendar month. The analysis is carried out for time periods starting on every trading day during the time of analysis. Results reported for a certain date are obtained from a period of 21 trading days ending at the stated date.

The correlation coefficient 2 is undefined if the logarithmic return of a stock is identically zero during a time period, i.e., the price of the stock is constant over the entire month. This occurs for approximately 10 stocks in every period, and these stocks are removed from the analysis for that period. Different stocks were removed in different periods.

The correlation coefficient $p_{ij}(\tau)$ ranges between $-1 \leq p_{ij}(\tau) \leq 1$. The higher the correlation between two stocks, the greater the degree of co-movement, with 1 as a perfect co-movement in the same direction, and -1 as a perfect opposite movement. The similarity between two stocks can equivalently, and more intuitively, be measured by a **distance** $d_{ij}(\tau)$ given by [5],

$$d_{ij}(\tau) = \sqrt{2\,(1 - p_{ij}(\tau))}. \quad (3)$$

The distance $d_{ij}(\tau)$ ranges from zero, for stocks whose prices are perfectly correlated, to a maximum of 2 for entirely unrelated stocks. They are suitable for use with network algorithms that require nonnegative edge

Category	Category Info	Number	Proportion
Finance	Banks and other financial institutions	487	19.8%
Technology	Hardware and software companies	403	16.4%
Consumer discretionary	Retail, consumer goods, and leisure	439	17.9%
Health care	Pharmaceuticals and health care services	675	27.5%
Other	Industrials, Energy, Utilities and Telecommunications ...	452	18.4%
Total		2456	100.0%

TABLE I. Distribution of various sectors in the data set.

weights, such as constructions of minimum spanning trees (MST).

Below, Fig. 1 shows the average distance from the complete network, averaged over all possible pairs of stocks as a function of time. The average distance decreased dramatically at the start of the Covid-19 pandemic in March 2020, but recovered quickly soon thereafter. This indicates a short-term increase in the overall correlation between all stocks in the US stock market, as nearly all stocks fell simultaneously at the onset of the pandemic.

C. Edge Reduction Techniques

The distance calculations of Section II B result in a complete stock network where any two stocks are connected by an edge, even if the distance between them is large, and the similarity between them might be considered negligible.

The completeness of the network hides the structure of the correlations between the stocks. For a network-theoretical analysis, it is essential to suppress unimportant edges between dissimilar stocks in the complete network, so that the hierarchical organization of the network in communities of similar stocks becomes visible. Several approaches have been proposed in the literature to remove unimportant edges and retain the most important edges in the network. Two of these are used in the current study: The Minimum Spanning Tree (MST), first used in this context by Mantegna [5], and threshold networks proposed by Boginski et al. [14].

Other notable filtering techniques include the Planar Maximally Filtered Graph (PMFG) by Tumminello et al. [12], and its more computationally efficient cousin, the Triangulated Maximally Filtered Graph (TMFG) [13]. These methods retain a fixed number of edges (specifically $3n - 6$) under planarity constraints, ensuring a sparse yet topologically rich representation of the market structure. However, because the number of edges in PMFG or TMFG remains constant over time, changes in network density cannot be observed. Therefore, threshold networks were deemed more suitable for this study, as they allow the number of edges to vary dynamically with the correlation structure of the market, so that structural changes during the Covid-19 period can be seen more clearly.

1. Minimum Spanning Tree (MST)

The MST is a tree sub-network that has the minimum possible total sum of edge weights given the constraint that any two nodes (stocks) are connected by a path in the MST. It follows from the minimality constraint that the MST does not contain any closed paths. The MST is an efficient way to filter out less important edges. It can be considered as the skeleton of the financial network. The connectivity ensures one total network component that is not split into fragmented parts. The MST was derived from the complete network using Kruskal's algorithm [29].

The seminal study by Mantegna [5] was the first to utilise the MST for the analysis of stock market data, using stocks as nodes and edges based on the correlation between the returns of 500 stocks within the S&P 500 index over a 2-year period (1994-1995). Various further studies used the MST (see, e.g., [6, 8–10, 12]). These studies typically observed that stocks were separated into sub-tree communities, with stocks from the same industry sectors close together in the MST. However, these communities are known to be unstable, as small changes to the correlations can cause large differences in the resulting communities [30].

Similar observations were made in the current study. The MST gave unstable results both in the identification of central network stocks and stock sectors, as well as in the description of the community structure, as these properties changed over time. These preliminary results are not presented in the paper. They led to the conclusion that the MST was not itself a suitable object of analysis. However, because it guarantees that the network remains connected, it remained a vital stepping stone toward our analysis of threshold networks.

Earlier studies have reported that the total edge weight of the MST shrinks dramatically as a result of the volatile effect of a crisis on the stock returns, due to increased correlation [10, 31]. This is consistent with the results in Fig. 1. An investment portfolio that invests on the outskirts of the MST was found to be efficient in reducing portfolio risk, consistent with results from the seminal Markowitz optimal portfolio construction method [10]. During periods of high volatility seen in economic crises, the MST changes from being more star-like in structure to being chain-like, making the investment opportunities

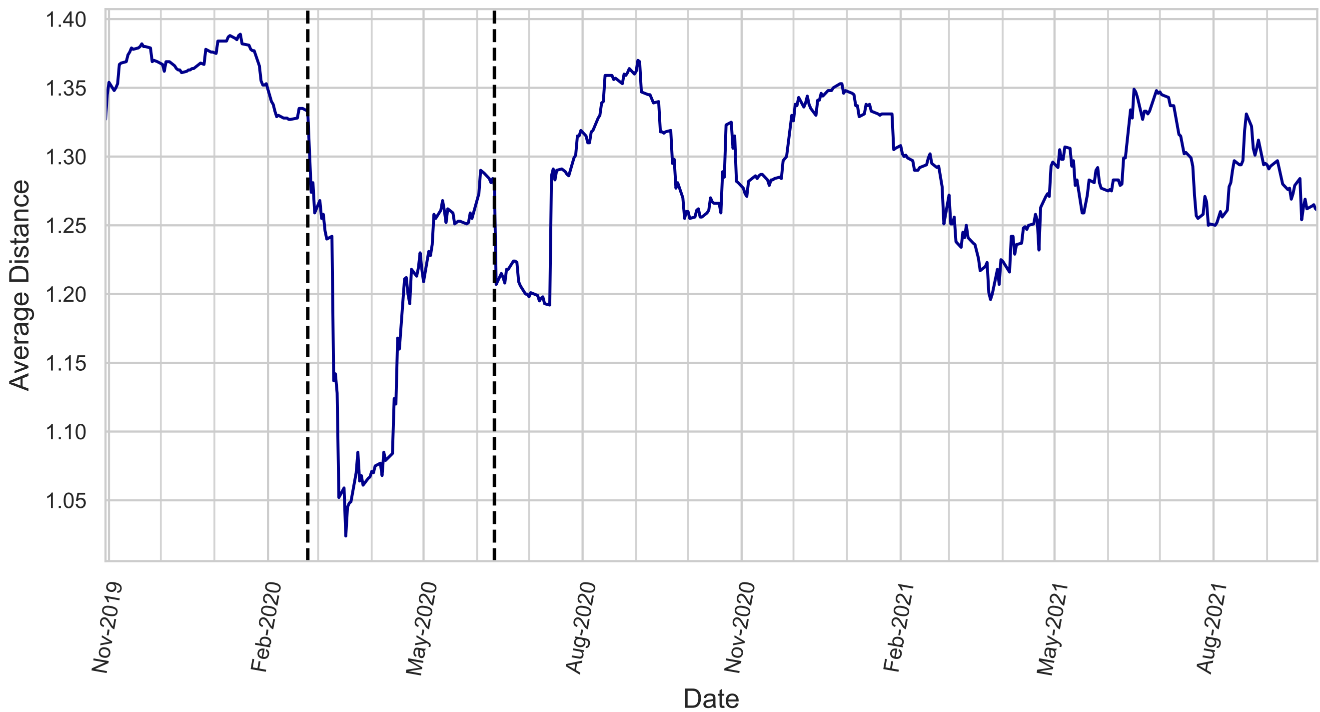

FIG. 1. Average distance in the network over time. The dotted black line highlights the 20th of February 2020, which was when the initial stock market crashed due to Covid-19 [28]. The second represents the worst single-day crash since the initial in March 2020 on 11th of June 2020 [28].

in the MST outskirts less diverse [32]. The recovery of tree length after a crash can aid as a signal of stock market recovery [32].

2. *Threshold Networks*

Tse et al. [15] argue that the MST and PMFG suffer substantial information loss because edges of high correlations are often removed, while edges of low correlations are retained just because they fit the topological reduction criteria. Due to these topological constraints, the resulting MST or PMFG may have unclear statistical and economic meaning [17]. A more informative reduction is provided by a **threshold network** [14], where all edges up to some maximum distance θ are retained and edges with a larger distance are discarded.

Rakib et al. [33] are among the few to observe the impact of a crisis on directed threshold networks. It was found that the most central companies during the Global Financial Crisis were in the sector of energy and financial services, while during the European Debt Crisis, the companies were in the communication services. Nobi et al. [34] constructed a threshold network and used a threshold value that was assigned based on the average and standard deviation of correlation coefficients. The authors considered the effects of the 2008 Global Financial Crisis on threshold networks of a local Korean financial market. During the crisis, it was reported that the network's correlation distribution had a higher spread of values, with fatter tails and lower kurtosis, resulting in a more densely connected network.

Few studies have been conducted analyzing Covid-19's effect on threshold networks, in contrast to the MST, and even fewer have observed the networks in terms of communities, central stocks, and attribute analysis. Wu et al [35] investigated the effect of Covid-19 on threshold networks. The authors analysed the closing prices of 228 stocks that belong to the Shanghai and Shenzhen 300 before and during the pandemic. Similar to other crises seen before, stock networks during the Covid-19 period are more densely connected. Heiberger et al. [22] built threshold networks to observe the dot-com and 2008 financial crisis, and saw that pre-crisis, the networks had well-partitioned communities based roughly on the sector, however as the crisis hit, the communities merged into bigger, less well-defined communities, due to the increased complexity of the network as the stocks becomes more similar.

The properties of a threshold network depend critically on the choice of the threshold θ. A common procedure in the literature is to compute the threshold network for various values of θ and then identify thresholds that lead to networks with desired characteristics [15, 16, 22, 24, 33–41]. However, this procedure is not practical if a temporal network is to be analysed at many different times. For this study, it is essential to have an algorithm that will identify a suitable threshold for each network, and that will allow a comparison of the results at different times. This remains a challenging problem [16].

Xu et al. [42] propose a method to find an optimal threshold that is applied to the threshold networks over the entire period of analysis. They use a consistence function to measure how the threshold networks change from one time step to the next. The threshold is chosen such that the changes in the threshold networks are maximally correlated with the changes in the original weighted network.

By contrast with the work of Xu et al. [42], we aim to find a threshold value separately for each time frame, so that the threshold is adapted to the individual network and makes its structure clearly visible. A suitable threshold should satisfy the following criteria:

1. The number of fragmented network components should be small.

2. The number of edges should be low.

As these criteria are in constant tension with each other, it is essential to find a compromise between them. Removing too few edges hinders community detection and obscures the potential for observing centrally influential stocks, while significantly increasing the runtime of network algorithms. On the other hand, excessive edge reduction may disconnect the network into separate subcomponents, leading to an unrepresentative analysis and data loss.

We propose to use the maximum value of weighted edges in the MST. This is the smallest threshold that ensures that the resulting threshold network is connected, with a single component. It will therefore produce the sparsest possible connected threshold network.

In a related approach, Ku et al. [41] depart from the requirement of a threshold network and instead include all edges of the MST as well as those that remain below a prescribed distance threshold. This construction guarantees a connected network and therefore satisfies the first of our criteria, but it leaves open the choice of threshold and is therefore not suitable for use in a temporal network with a large number of time frames.

Figs. 2 to 4 illustrate the performance of the suggested threshold for a network that captures the first 21 trading days of November 2019. Fig. 2 shows that the number of components in the threshold network rises rapidly if a lower threshold is chosen. Conversely, a larger threshold leads to a network with many more edges, as seen in Fig. 3. Indeed, the proposed threshold filters out more than 95% of edges from the complete network, agreeing with the edge sparsity criterion. It remains to see how well the threshold network captures the community structure of the market. This question will be addressed in Section III.

Although the threshold selection method is illustrated here using data from October 2019, the threshold is not constant over time. It is recalculated for each temporal network from its MST. This ensures that the threshold is appropriate for every network and removes any bias that might be introduced by fixing a global threshold. However, the variation of the threshold is small. The threshold is always close to 0.98, with a standard deviation of approximately 0.03. Moreover, the threshold method was also evaluated, by the same criteria used here, for data after October 2021, outside the period of analysis. The results are similar to those shown.

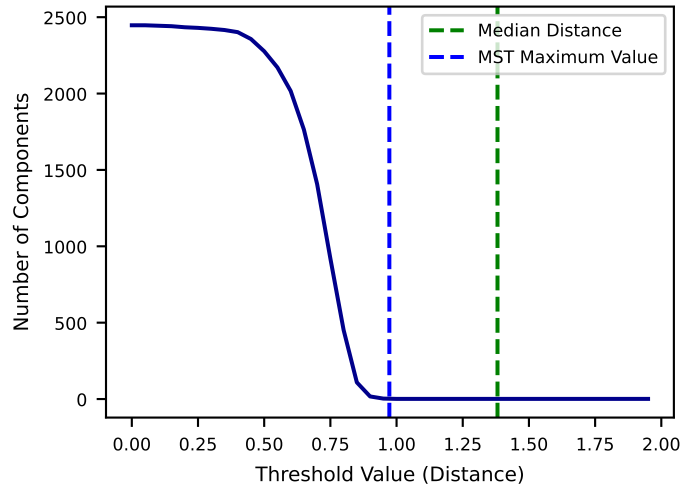

FIG. 2. Number of components in the threshold network for November 2019 for varying threshold. The proposed threshold produces a connected network. For smaller thresholds, the number of components increases rapidly.

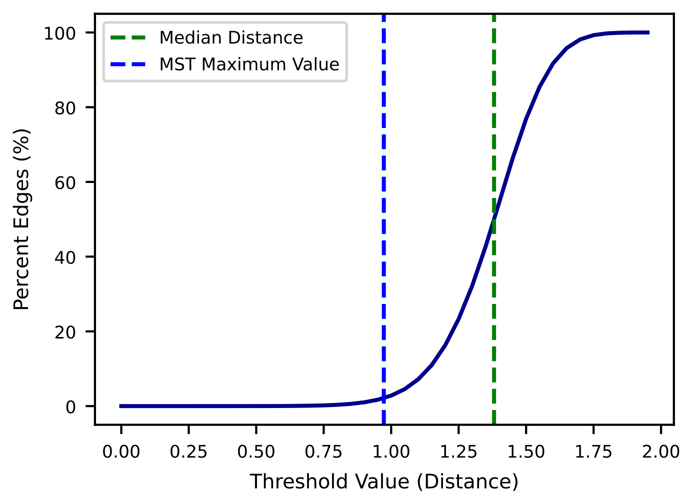

FIG. 3. Percentage of edges remaining of the threshold networks for November 2019 for varying thresholds. The proposed threshold removes a large percentage of edges.

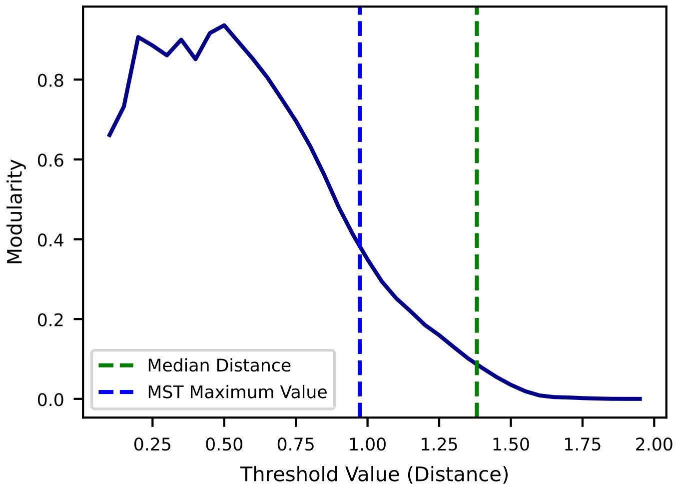

FIG. 4. Modularity of the threshold for November 2019 for varying threshold. The proposed threshold significantly beats the minimum requirements of modularity at 0.3.

III. COVID-19 IMPACT ON COMMUNITY STRUCTURE

A **community** of stocks in a network is, loosely speaking, a group of nodes that are more closely related to each other than to the rest of the network. Once a network has tentatively been subdivided into communities, how appropriate this subdivision is can be measured by the modularity [43]:

$$Q = \sum_{c=1}^{n} \left(\frac{L_c}{m} - \left[\frac{k_c}{2m} \right]^2 \right),$$

where m is the total number of edges in the network. The sum iterates through all communities c, k_c is the sum of all degrees within community c and L_c is the number of edges between a node in community c and a node outside it. The modularity measures the density of edges within the proposed community compared to edges that connected different communities. The larger the modularity Q, the better the the proposed subdivision reflects the actual community structure of the network. A modularity of 0.3 or above is commonly considered a good indicator of significant community structure in a network [44]. We used the **Louvain algorithm** proposed by Blondel et al. [45] to identify a subdivision into communities that approximately maximises the modularity.

Fig. 4 shows how the modularity of the threshold network depends on the chosen threshold for the data of November 2019, the same as in Figs. 2 and 3. For the proposed threshold, the maximum edge weight in the MST, the modularity lies well above the guideline of 0.3. Much higher values of modularity could be achieved using lower threshold values. However, in a network that is extremely sparse, there is a risk that the community structure might be an artefact of the edge reduction. One may wish to sacrifice the connectedness of the network for higher modularity if one's goal is only to observe community structure, by lowering the threshold slightly below the maximum edge weight of the MST. For the period in Fig. 4, the impact would be negligible. Overall, we conclude that the maximum edge weight of the MST provides a suitable threshold that balances the three desired criteria and can easily be applied to the construction of a large number of networks.

The number of communities returned by the Louvain algorithm depends on the network and therefore varies over time. For most time frames, our networks are decomposed into approximately 7 communities. An example of the community structure is shown in Fig. 5. In this time frame, the Louvain algorithm yields 11 communities, somewhat above average. The most dense community is indicated in pink, which consists mainly of financial stocks.

Fig. 6 shows the time evolution of the modularity in the temporal threshold networks. The Covid-19 impact on modularity is drastic. The volatile dip in prices during the onset of Covid-19 weakened the community structure for several months, until an eventual recovery in August 2020. Modularity dipped significantly during this period, reaching a minimum at the peak of the crash. Moreover, there was a small drop in modularity a month prior to the volatile drop in prices in March 2020, which may have been due to the anticipation of Covid-19 impacting the markets before investors could realise its severity.

Because at the onset of the pandemic all stocks are highly correlated (see Fig. 1), the number of edges in the threshold networks is high. This will be discussed in detail in Section IV. As a consequence, the community structure of the network is less pronounced, and the modularity is low. Pre-Covid-19 the communities were stable and well-defined with high modularity, each being affected by its own unique factors that influenced the structure of the community, be it sector industry factors or geographic factors. These different factors differentiate the communities. But suddenly, an overarching huge significant factor, like Covid-19, vastly affects the entire market. All the stocks are affected by one factor, which is so strong, it overrides all the other smaller factors, compressing the network into denser communities, deemed insignificant due to the vast propensity of increased edges due to an increased correlation, which decreases distance (see Fig. 2).

For a deeper understanding of the community structure, it would be interesting to see how the composition of communities changes over time, particularly with respect to sector. However, as the number of communities fluctuates, it is difficult to track the same community over time. For this reason, we focus on two communities in each period, the largest community and the densest, which are easily identified. The largest community is defined as the community with the most stocks and is, at most times, well separated from the smaller communities. The most dense community is the community that

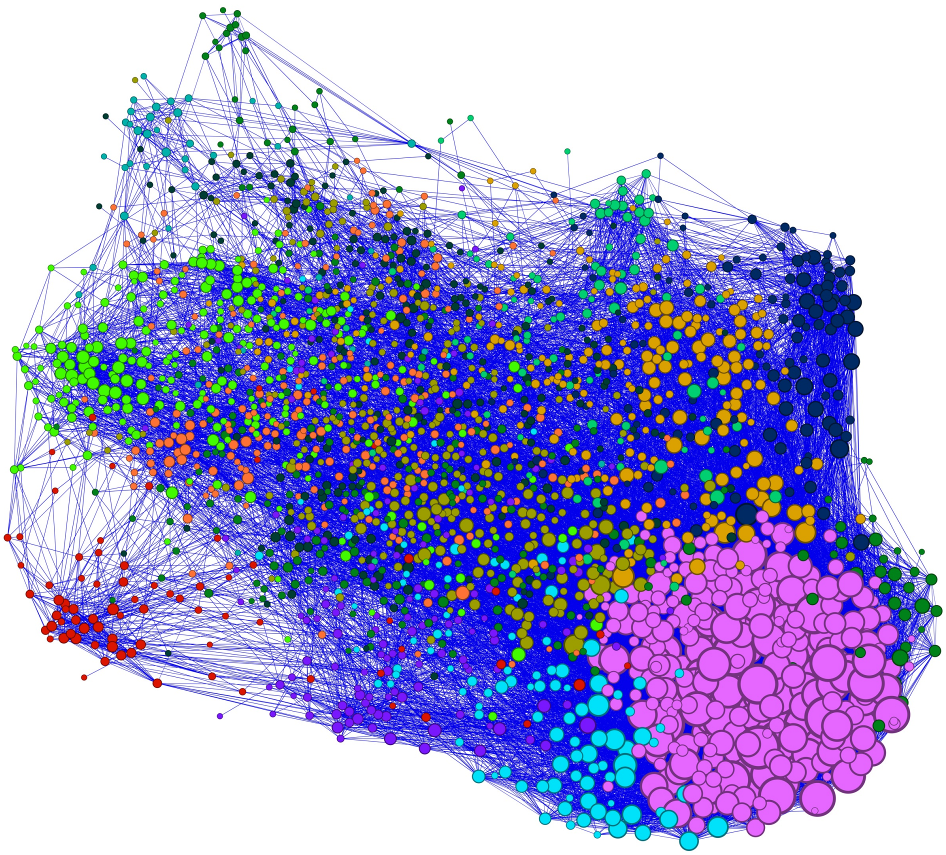

FIG. 5. Threshold network of 2456 stocks using the first 21 days of November 2019, highlighting 11 communities (colours) formed using the Louvain algorithm. The dense pink community consists mainly of financial stocks (50-60% composition). The larger the stock, the higher the number of connections.

has the highest density of edges. Density is calculated by taking the number of edges divided by the number of edges in a complete network of that size. The size of the most dense community is restricted to be 50 at a minimum, to avoid analysing communities that are too small.

Fig. 7 shows the composition of the largest community over time. This should be compared to the data in Table I, which shows the sectoral composition of the underlying data set. In the data set, the five sectors contribute roughly equal numbers of stocks (though the health care sector is somewhat overrepresented with 27.5% of stocks). By contrast, the largest community is, in most time frames, dominated by health care and technology stocks. This is consistent with earlier findings [20, 46, 47], that stocks of the same sector are more likely to be in the same community.

At the onset of Covid-19, the composition of the largest community changed drastically, now containing an almost even split of the five sectors, with no clearly dominant sector. This change appears in the analysis with a delay of a few weeks because the correlation networks are constructed from 21 trading days or about a month's worth of data in the past of the date for which it is reported. Immediately after the onset of the pandemic, the network was still dominated by earlier data, leading to the delay that appears in Fig. 7. Once the time frame is dominated by pandemic-period data, all sectors contribute equally to the largest community. This effect persists for almost two months until May 2020. It indicates that during this period, the most important influence on market behaviour, the impact of the pandemic, was the same for stocks in all sectors, whereas at other times, sector-specific factors play an important role. These findings are consistent with those of Heiberger et al. [22], where the community structure based on sector collapsed during crisis.

Financial stocks are found in most time frames to contribute the least to the largest community. They are underrepresented there, compared to the composition of the overall network, because they strongly dominate a different community, namely, the most dense community, whose composition is displayed in Fig. 8. Conversely, health care and technology stocks that dominate the largest community are underrepresented in the most dense community. When Covid-19 hits the stock market, the most dense community ceases to be dominated by financial stocks and instead contains equal numbers of stocks from all five sectors.

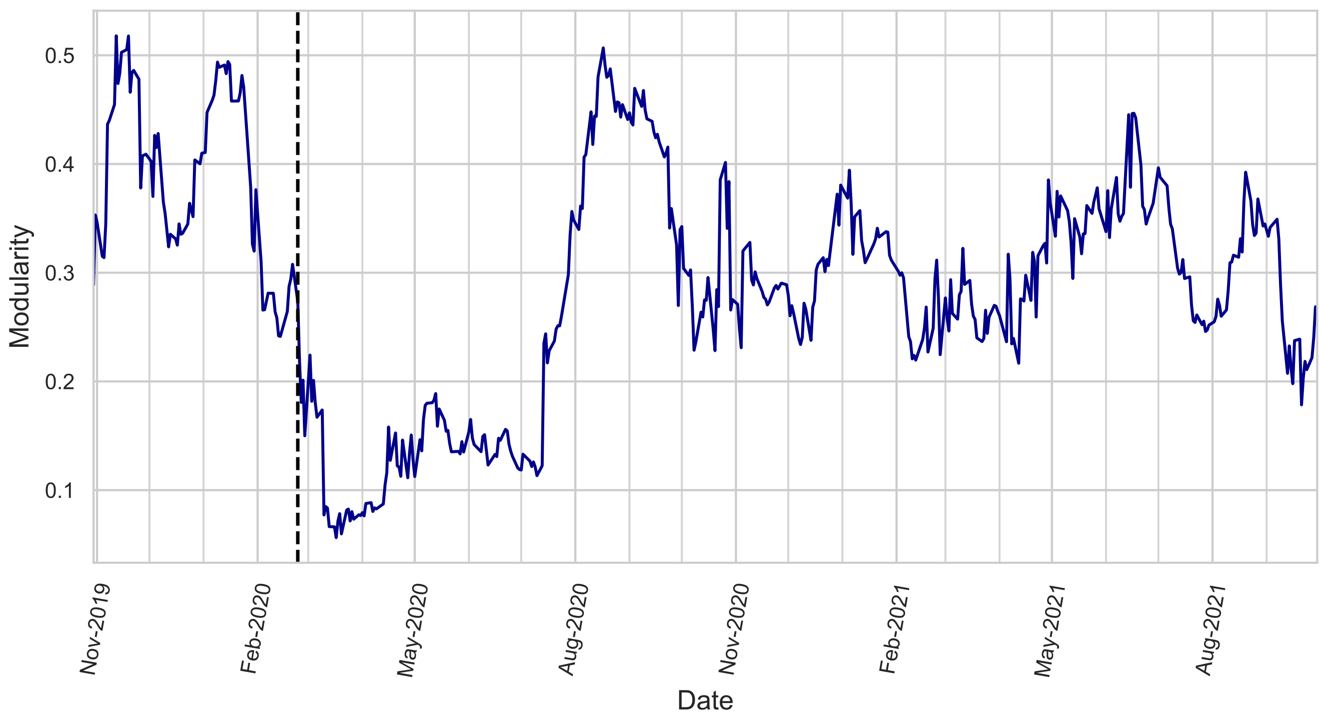

FIG. 6. Modularity of each threshold network over time. The dotted black line highlights the 20th of February 2020, which was when the initial stock market crashed due to Covid-19 [28].

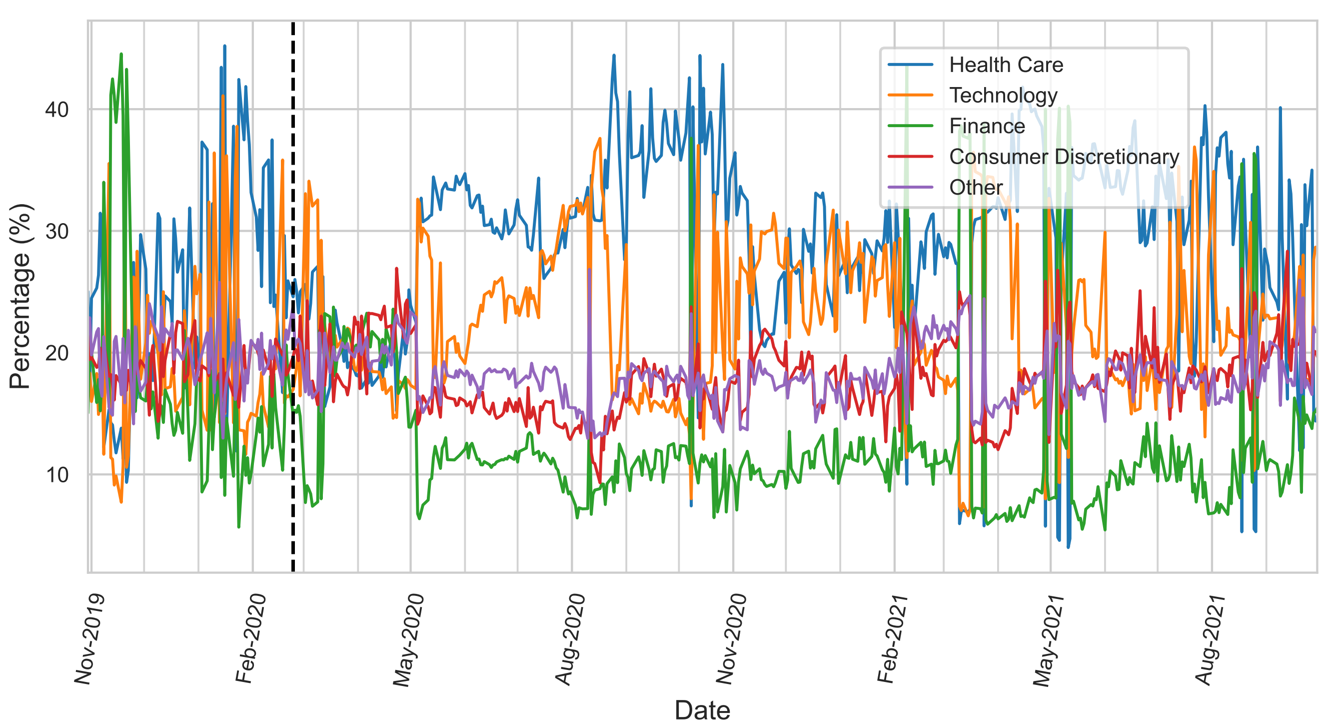

FIG. 7. Percentage of each stock sector within the largest community over time.

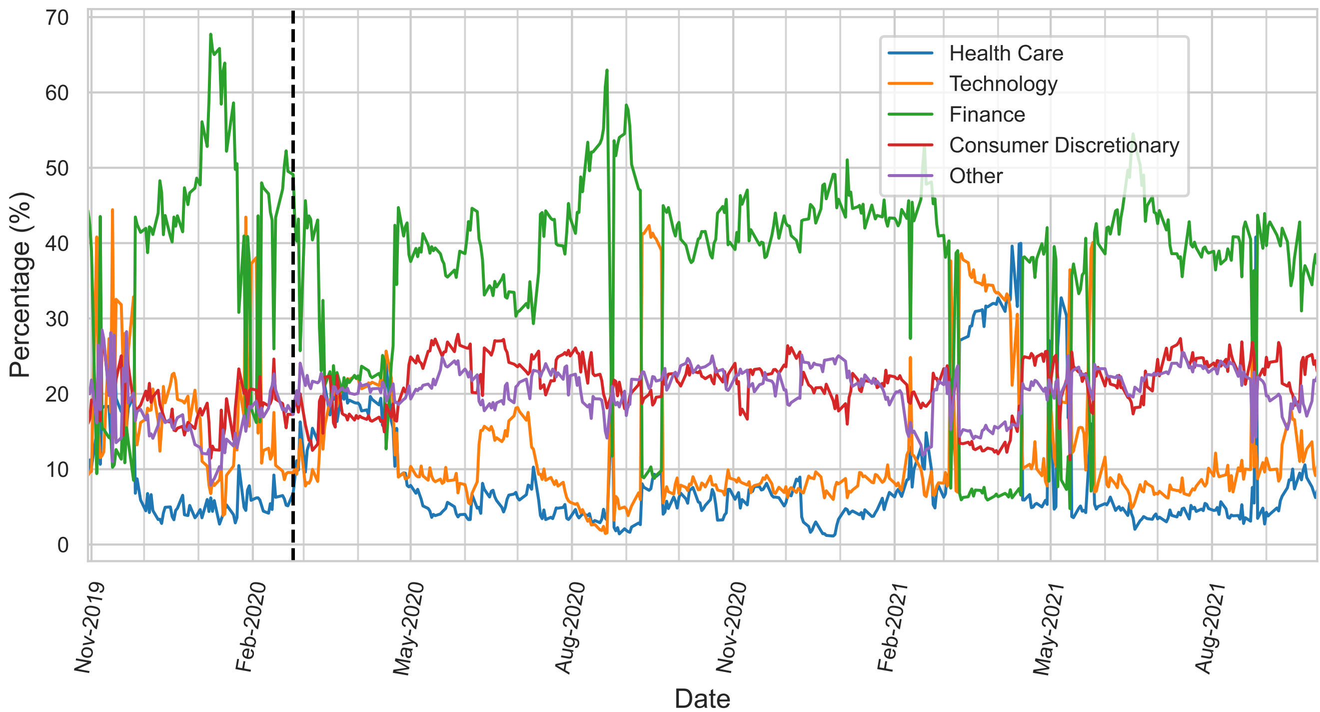

FIG. 8. Percentage of each stock sector within the most dense community over time.

IV. HIERARCHICAL STOCK CENTRALITY

After analysing the community structure of the networks, we proceed to identify the most central stocks. The centrality of a node is here measured by its degree, i.e., by the number of edges it has. A similar analysis was carried out for closeness centrality, another common measure of centrality [48]. The findings agree with the results obtained from degree centrality and are not presented here.

Fig. 9 shows the average degree of all stocks in a particular sector and Fig. 10 shows the corresponding data for stocks in a particular volatility class. To determine the volatility classes, the standard deviation of logarithmic returns was computed for every stock over the study period. Stocks were split into three classes with equal numbers of stocks, corresponding to the top third, middle third, and bottom third of the distribution of volatility.

The impact of Covid-19 is vividly clear. In the months before Covid-19, the average degrees were comparatively low and vary only a little, reflecting a period of stability. They surge in March 2020, which signifies that the network becomes extremely dense. This spike is driven by a drop in average distance between stocks because of increased correlation between them due to a sharp, volatile price drop (see Fig. 1). Markets partially recovered in April and May 2020, but a second spike occurred in June and July 2020. It reflects the market shock of 11/12 June, where 90% of stocks in the market fell sharply as a result of fears of a second Covid-19 wave, resulting in the worst sell-off since March 2020 [28]. The second spike appears with a width of approximately one month, reflecting the duration of the time window of 21 trading days used in the analysis. Average degrees return to pre-pandemic levels for a short period in August 2020, although smaller disturbances remain common throughout the period of analysis. The variation of average degrees over time is markedly higher after the Covid shocks than it was before.

Pre-pandemic, the stocks with the highest average degrees were those in the finance sector, as well as stocks of low volatility. They remain central during and after the Covid-19 crisis and show the highest peaks. Stocks within these sectors became most tightly connected relative to the others, even more so than before the crash. Investors monitor the performance of large market cap stocks, particularly in the technology industry (Apple, Google, etc.) and large banks, as a benchmark for the overall performance of the market. Generally, technology contains the stocks of well-known and established companies, while large banks are highly sensitive to the economic environment, with changing interest rates and economic policy [46]. If these perform poorly, investors are prompted to panic sell their stocks in other investments, making the overall market move with these stocks. Hence, stocks are more likely to be correlated with these major stocks and be more central in the network. During this time, International Bancshares Corp (IBOC) was the most central in the network with a degree of approx-

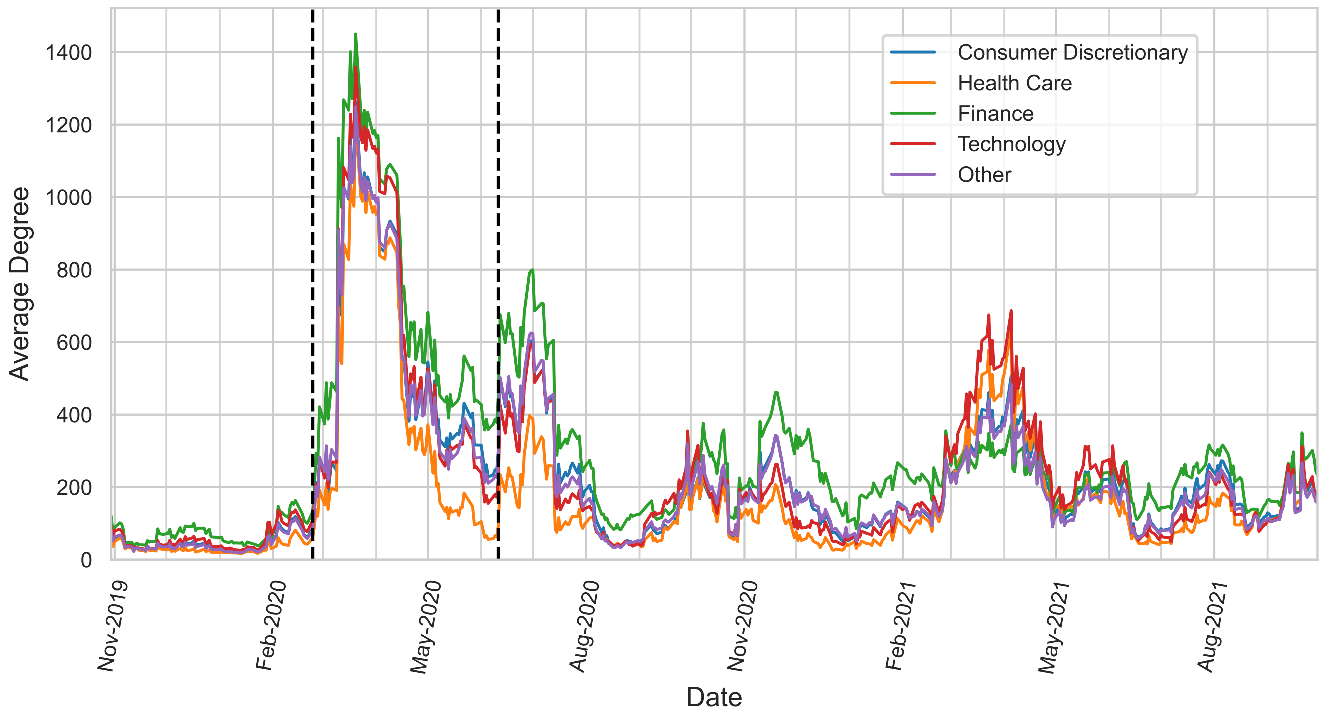

FIG. 9. Average degree over time, split into different stock sectors. The dotted black line highlights the 20th of February 2020, which was when the initial stock market crashed due to Covid-19 [28]. The second represents the worst single-day crash since the initial in March 2020 on 11th of June 2020 [28].

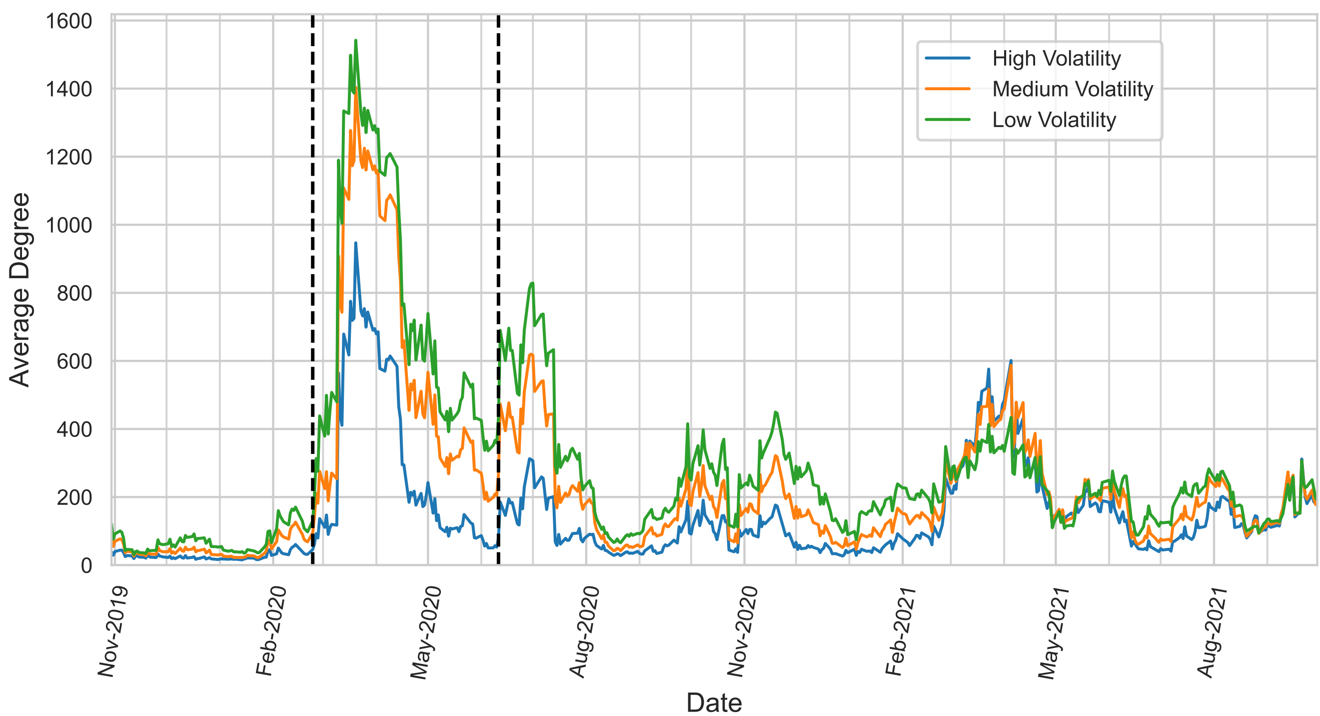

FIG. 10. Average degree over time, split into different volatility levels.

imately 1600.

Health care stocks showed a relatively lower average degree post-crash, reflecting more independent behaviour, likely because the sector benefited from pandemic-driven demand rather than broad market fear (see Fig 9). Health care stocks also usually have higher volatility. Investors preferred to avoid risky, volatile investments during Covid-19, which is consistent with potentially why the degree of highly volatile stocks became significantly lower relative to their lower volatility counterparts.

An unusual disturbance of the market occurred in March 2021. It once again led to anomalously dense networks. In this case, technology stocks and stocks of high volatility became more central, whereas the effect on financial stocks and stocks of low volatility is smaller. In this respect, the anomaly of March 2021 differs markedly from those caused by Covid-19. Its cause is open to speculation. There may be links to the GameStop (GME) technology stock short squeeze and cryptocurrency price explosions [49].

V. CONCLUDING REMARKS

The financial markets are an enormously complex, interrelated system. Network science provides a powerful method to model its structure and observe its dynamics over time [1]. This study applied these tools to a large data set of stock logarithmic returns, analyzing the entire NASDAQ stock exchange for a two-year period from November 2019 to October 2021. The correlation between the logarithmic return of stock price movements was computed for every pair of stocks over a moving time-interval of a month. A time-varying threshold network was constructed in which edges were retained only between sufficiently highly correlated stocks. A technical hurdle in this analysis is the identification of a suitable threshold that can be easily and efficiently computed for a large number of networks and that yields threshold networks that clearly reflect the structure of the markets. It was shown that the maximum edge distance (minimum correlation) that appears in the Minimum Spanning Tree (MST), satisfied these requirements. This prescription is independent of the nature of the networks that are being analyzed and can easily be transferred to other applications in network science.

The onset of the Covid-19 pandemic drastically reshaped the community structure of the threshold network. The modularity of the network significantly decreased, and previously well-separated communities collapsed into less well-defined ones because of the overwhelming influence of the pandemic on most stocks. Correspondingly, an analysis of central stocks in the network revealed that the Covid-19 crisis triggered a temporary collapse in market diversification, unifying all sectors, and volatility groups into a single, highly interconnected structure. Average degree surged sharply in March 2020 and again briefly in June 2020, during the second major sell-off, reflecting extreme market co-movement and systemic risk. Notably, the pandemic disrupted the usual dominance of certain sectors in the financial network, resulting for a short time in a similar performance of otherwise disparate stocks. As a consequence, available investment opportunities were less diverse. As volatility subsided in the aftermath of the pandemic, the network gradually re-established its previous hierarchical structure, with sectoral distinctions and relative independence of the sectors resuming by late 2020. These highlight the value of network-based approaches for monitoring and assessing systemic market stress and indicators of recovery.

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